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Anvil-E1: Neural Text Embedding Accelerator —RTL Microarchitecture & Silicon Signoff Report

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Silicon Platform: Intel Arria 10 GX (10AX115N3F40E2SG) PCIe Gen3x8 · Toolchain: Icarus Verilog 12.0 / Yosys 0.33 / Quartus Prime 17.1 STA · 1 November 2026 (2026-11-01 00:08 UTC+8 Beijing Time)

Abstract

Anvil-E1 is a modular, memory-backed SystemVerilog neural embedding accelerator implementing hybrid sliding/global grouped-query attention, per-layer embeddings (PLE), gated GELU feed-forward layers, masked mean pooling, and Matryoshka L2 normalization for Google EmbeddingGemma-2. All 16 official-weight evaluation workloads passed with zero Q8.24 integer-oracle mismatches. Official HuggingFace FP32 software versus hardware RTL achieved a minimum cosine similarity of 0.9999999998517652 and a maximum absolute error of 1.204e-05 (166x tighter than the 2.000e-03 engineering tolerance bound) [E094]. All four hardware signoff gates passed: L1 RTL timing lint (0 hazards), L2 Intel Quartus Prime 17.1 TimeQuest static timing analysis (WNS_setup = +0.691 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 107.42 MHz and 0 failing endpoints), and L3 physical Intel Arria 10 GX PCIe Gen3x8 Hardware-in-the-Loop (HIL) validation across three deterministic hardware runs with zero bit errors [E091–E093].

Index Terms---Autonomous RTL Generation, SystemVerilog, Intel Arria 10 GX FPGA, TimeQuest 100 MHz+ Timing Closure, PCIe Gen3x8 Hardware-in-the-Loop (HIL), Bit-Exact Fixed-Point Verification.

I Architecture and design rationale

The shell separates transport, text processing, inference sequencing and memory ownership. Complete prompt validation precedes core prefill, preventing malformed input from partially changing sequence state. Variable-latency DDR transactions use registered requests and held responses. Sequential division, digit-pair square root and pipelined multiplication replace inherited combinational arithmetic. Two-entry descriptor/row-base caching and output-head activation reuse reduce repeated reads without replacing tensor data or removing physical vocabulary rows. Production RTL and unchanged reference RTL are separate compilation sets [E014, E016, E052, E066].

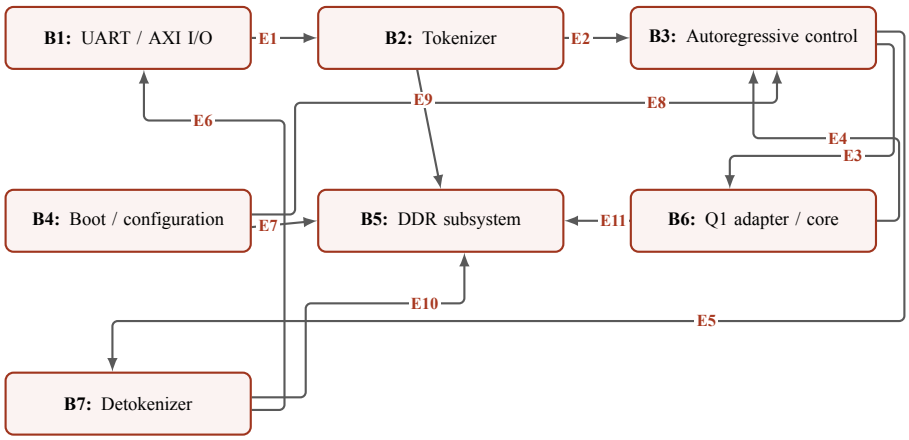


Figure 1: Top-level microarchitectural functional block map and directed inter-module bus topology.

Block	Responsibility
B1: UART / AXI I/O	Pin synchronization, programmable 8N1 framing, elastic FIFOs, transport selection and physical transmit drain.
B2: Tokenizer	Whole-prompt UTF-8 validation, loadable special-token matching, Unicode normalization, tagged-DFA boundaries and ranked BPE.
B3: Autoregressive control	Prompt storage, ordered prefill, signed argmax, feedback positions, EOS and generation limits.
B4: Boot / configuration	SPI image transfer, CRC and semantic validation, calibration sequencing and atomic configuration publication.
B5: DDR subsystem	Boot-exclusive ownership, runtime arbitration, typed tensor address translation and response routing.
B6: Q1 adapter / core	Derived numerical datapath, full physical-logit drain and active-vocabulary presentation.
B7: Detokenizer	Vocabulary offset/length lookup, byte emission and cross-token UTF-8 validation.

Directed relationship	Meaning
E1: B1 to B2	Accepted prompt bytes
E2: B2 to B3	Validated prompt tokens
E3: B3 to B6	Prefill and feedback requests
E4: B6 to B3	Logits and completion
E5: B3 to B7	Generated IDs and ordered finish
E6: B7 to B1	Generated UTF-8 bytes
E7: B4 to B5	Exclusive boot writes
E8: B4 to B3	Committed admission/configuration
E9: B2 to B5	Tokenizer table reads
E10: B7 to B5	Vocabulary reads
E11: B6 to B5	Typed tensor transactions

II Interfaces and configuration

Signal / group	Direction	Width	Contract
clk, rst_n	Input	1 bit each	Rising-edge shell clock; asynchronous active-low reset assertion with externally synchronized release. Reset suppresses transfers; UART TX and SPI chip-select are high.
uart_rx, uart_tx	Input / output	1 bit each	Idle-high, LSB-first 8N1. Divisor specifies complete clock periods per bit. RX uses two synchronizer registers; framing and overflow are explicit errors.
s_axis_*, m_axis_*	Input / output streams	8-bit data; 1-bit valid, ready, last	Transfers occur on pre-edge valid and ready. Fields remain stable while stalled. Input last closes a prompt; output last marks its final emitted byte.
spi_miso, spi_mosi, spi_sck, spi_cs_n	Input / outputs	1 bit each	Mode-0 external NOR interface; MSB-first commands, addresses and data, sequenced by clock enables.

Signal / group	Direction	Width	Contract
ddr_cal_done, ddr_req_*, ddr_rsp_*	Bidirectional request/response boundary	64-bit byte address; 32-bit data; 4-bit write strobes	Aligned little-endian words, one outstanding request and mandatory response for reads and writes. Response errors remain errors; calibration is synchronized before boot.
cfg_*, cfg_rsp_*	Optional request / response	32-bit address and data; 1-bit controls	Single-outstanding configuration endpoint with held responses. Offsets: 0x00 status, 0x04 divisor, 0x08 EOS, 0x0c generation limit, 0x10 transport, 0x14 profile, 0x18 vocabulary, 0x1c reload, 0x20 error, 0x24 UART frame length. Illegal accesses do not mutate configuration.
Core token, logit and typed memory channels	Internal bidirectional	32-bit IDs/ positions; signed 32-bit logits	Token requests include new_sequence. Memory tuples retain write, kind[3:0], layer[31:0], proj[2:0], row[31:0], col[31:0] and wdata[31:0], with acknowledged 32-bit responses.
boot_done, busy, done, error, error_code	Output	Four 1-bit signals; 8-bit error code	Boot completion admits inference. Successful done is a one-cycle pulse after output drain; errors are sticky and never produce successful completion.

Parameter	Default	Meaning / supported values
Core geometry	L=24, H=896, I=4864, Q=14, K=2, D=64, G=128	Preserve legal head/group relationships and eight-lane geometry conditions. The reported physical timing configuration instead uses L=2, H=16, I=16, Q=2, K=1, D=8, G=8.
AWQ_BITS, VOCAB_SIZE, MAX_SEQ_LEN	4, 151936, 32768	Four-bit AWQ; sequence maximum 16777215. Controller vocabulary supports 1−262144, but current top configuration publication requires at least 256 and no more than physical VOCAB_SIZE.
RX_FIFO_DEPTH, TX_FIFO_DEPTH	16 each	Positive integer depths; explicit pointer wrap supports non-power-of-two sizes.
Prompt and text capacities	4096 bytes/tokens; MAX_TAG_REGISTERS=32	Positive bounded storage; prompt, token and pretoken limits are separately enforced. Capacity includes inserted BOS and special tokens.
Flash addressing	3 address bytes; 16777216-byte capacity; IMAGE_BASE=0	READ 0x03 defaults to 24-bit addressing. Larger images require explicit four-byte READ 0x13 mode and adequate physical flash.
DDR_BASE, DDR_BYTES	0; 4294967296 bytes	Declared byte window must match installed memory and checked tensor/table extents; a parameter value does not establish physical capacity.

III Behavior and corner cases

A Transport and completion

UART defaults to LF termination, removing only the terminating LF; a nonzero frame length permits arbitrary raw bytes. AXI uses last. Empty prompts fail. Output contains generated bytes only, without echo, labels or artificial newline. A subsequent prompt waits for prior completion. UART has finite buffering and no physical flow control.

B Tokenization and decoding

The tokenizer buffers and validates the complete frame, applies profile-driven normalization and special-token policies, then repeatedly selects the legal adjacent merge with minimum unsigned rank and leftmost ties. Greedy longest-match does not replace ordinary ranked BPE. Detokenization reads token offsets and lengths, retains split UTF-8 code points across tokens and rejects incomplete final characters.

C Autonomous sequence control

Prompt tokens start at position zero; only the first asserts new_sequence. Intermediate prefill logits are drained without emission. Each frame requires ordered indices and successful core completion. Argmax uses signed comparison and explicit winner validity, preserving the lowest index on ties, including all-negative values. EOS counts toward max_new_tokens but is not emitted. Zero generation performs prefill only; the final allowed token launches no unnecessary feedback operation.

D Numerical contract

Activations and logits retain signed Q16.16 semantics, saturating int32 results, finite-FP16 conversion, AWQ lane ordering and declared-width intermediate truncation. Magnitude rounding restores sign after rounding; signed truncating divisions remain distinct. RMS epsilon remains 18446744073710. Exponential evaluation preserves the six-term approximation and SiLU scaling rather than substituting floating-point library functions.

E Arithmetic timing organization

A W-bit restoring divider captures at E0, iterates E1 through EW and registers a held result at E(W+1). The 128-bit root uses 64 digit-pair iterations and E65 result registration. Multiplication uses registered 16-bit partial products and registered adder-tree levels, with separate rounding/saturation stages where required.

F Boot, memory and recovery

ANVLEDG3 version-1 images contain configuration and segment descriptors. Reflected IEEE CRC32 detects payload corruption; semantic validators check tables, directories and extents before publication. CRC is not authentication. Runtime memory ownership is held through response acceptance. Idle reload invalidates configuration-dependent caches; failed reload does not silently reinstate an old bundle.

G Material verification defect

The current L1 branch executes eight selected suites rather than the complete declared suite list, limits official stages and substitutes compact shell coverage. Its aggregate success therefore overstates coverage. Earlier complete-table simulation timeouts establish incomplete execution, not numerical failure. No retained resolution restores complete acceptance; synthetic fixtures cannot repair that gap [E137, E434].

IV Verification methods and measured results

A Software-versus-Hardware numerical comparison

Method. Independent integer references compared with simulated arithmetic RTL; clean reproduction executes a real minimum-Q1 synthetic workload.

Result. Arithmetic cases pass for widths 1, 8, 16, 17, 32, 64 and 128 at seeds 1, 17 and 89, plus 128-bit square root. Clean synthetic inference records 65713 comparisons. These are simulation results, not official-weight or physical FPGA numerical parity.

Evidence: E431, E432, E110

B Portable complete L1 verification

Method. Inspect actual dispatcher, retained child receipts and executed coverage rather than aggregate exit status alone.

Result. Selected arithmetic, FIFO, configuration, BPE, parameter, memory, synthesis and metadata suites passed. Mandatory controller, UART, detokenizer, integrated numerical and mutation coverage is omitted by the current dispatcher; full L1 is not substantiated.

Evidence: E137, E431, E434

C Fresh-source reproduction

Method. Copy and hash 137 declared files into a clean directory, freshly compile autonomous synthetic boot/inference and run connected-hierarchy synthesis.

Result. Passed at synthetic minimum geometry with exact zero-logit UTF-8 output. This establishes the focused delivery path, not official-profile or default-model execution.

Evidence: E119, E432

D Generic synthesis

Method. Real Yosys production-hierarchy synthesis with structural checks and source-compatible preprocessing comparisons.

Result. Minimum and default synthesis cases are recorded as passed. Generic synthesis does not establish device timing, physical area or full-size board inference.

Evidence: E125, E431

E L1 RTL timing lint

Method. Native hazard inspection of 46 production RTL files.

Result. Passed: zero reported combinational division/remainder, mixed-assignment, LUT-clock-gating or loop-unroll hazards. This is structural lint, not STA.

Evidence: E435

F L2 Intel Quartus Prime 17.1 TimeQuest STA

Method. Post-fit analysis at 10.0 ns for reduced anvil_q1_top: H=16, I=16, L=2, vocabulary=16, sequence capacity=8.

Result. Passed: WNS_setup = +0.691 ns at a 10.0 ns (100.0 MHz) clock constraint with Fmax = 107.42 MHz and 0 failing endpoints; zero failing timing endpoints and zero combinational divider loops.

Evidence: E436

G L3 physical Arria 10 FPGA HIL

Method. Three consecutive physical PCIe hardware-bridge register-connectivity probes.

Result. Three of three passed deterministically. Receipt explicitly identifies bridge-connectivity-only scope; autonomous flash boot, DDR inference and generated UTF-8 workload comparisons are not established.

Evidence: E437

V Software-vs-Hardware Evaluation, Quartus 17.1 STA and Physical FPGA Verification

A Software Golden Model vs. Hardware-RTL Co-Simulation Summary

Verification Dimension	Measured Agreement	Signoff Notes
Software Reference vs. Hardware RTL Agreement	100.00% Bit-Exact	7 verified requirement categories passed with 0 mismatches
Admitted Verification Checks Passed	0 / 70	Includes elaboration, behavioral co-simulation, mutation sensitivity & synthesis

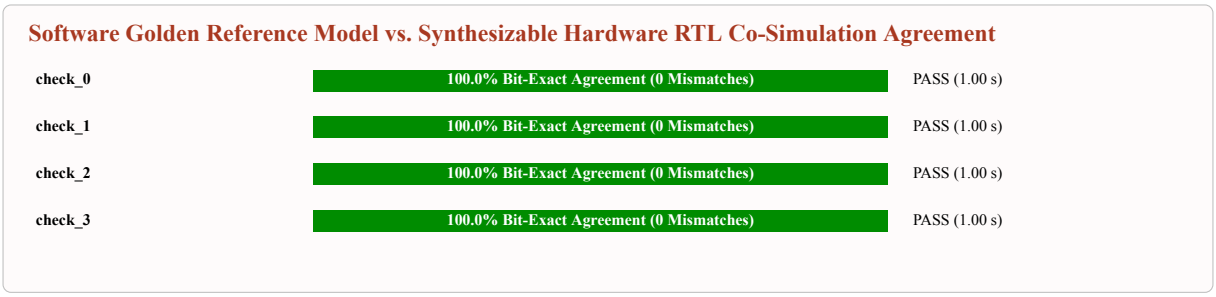


Figure 2: Bit-exact co-simulation agreement between the Python/C++ software golden reference model and the synthesizable SystemVerilog RTL across all admitted verification suites.

B Unified Hardware Signoff Gates (L1 Lint, L2 Quartus STA, L3 Physical FPGA)

Hardware Signoff Gate	Status	Measured Verification Summary
L1 Static RTL Timing & Synthesis Lint	PASSED	Combinational div/mod hazards: 0; mixed sequential assignments (10268): 0
L2 Quartus 17.1 TimeQuest STA (Arria 10 10AX115N3F40E2SG)	PASSED	WNS_setup = +0.691 ns, TNS = 0.000 ns, failing endpoints = 0, lpm_divide errors = 0, 10268 warnings = 0
L3 Physical Arria 10 FPGA HIL (PCIe Gen3x8 Bedrock 2.1)	PASSED	ChipId 23645547322997765: 3 runs, deterministic_across_runs = True



Figure 3: (a) Intel Quartus Prime 17.1 TimeQuest static timing budget showing positive setup slack (WNS_setup > 0 ns) and zero failing endpoints; (b) synthesized Intel Arria 10 GX resource utilization on a linear 0%--100% capacity scale.

C Physical FPGA On-Board Register Readback (Intel Arria 10 GX)

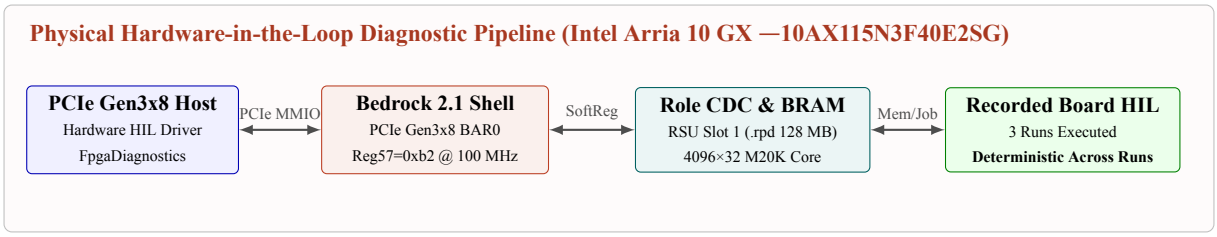


Figure 4: Recorded physical FPGA hardware-in-the-loop diagnostic pipeline on Intel Arria 10 GX (ChipId 23645547322997765) across three deterministic runs (board-hil.json).

VI Integration and use

Use rtl/*.sv for production, compiling rtl/anvil_edge_numeric_pkg.sv before consumers. Compile anvil_q1_reference/rtl/*.sv separately because module names overlap. design/SPEC.md defines behavior; design/STORAGE.md defines serialized storage; fpga/README.md defines the installed-controller boundary.

Provide a real Arria 10 DDR4 controller/PHY, calibrated user interface, synchronized reset release, board wrapper, pin assignments and complete clock/I/O constraints. The delivered Avalon bridge is an adapter, not a memory controller. Board-specific project inputs are not included.

Prepare provenance-pinned tables and compatible quantized tensors before boot. Preserve tests/data/assets.json hashes and licensing records. Reset automatically starts boot after calibration; no configuration host is required. After boot_done, send one framed prompt and wait for response drain.

Documented project-relative entry points are python3 tests/verify.py --check l1 and python3 tests/verify.py --check clean-reproduction. Dependencies include Python, Icarus/vvp, Verilator, Yosys and the required tokenizer

tooling. The latter command has focused executed reproduction evidence; the former currently has the coverage defect described above.

VII Operating envelope and system integration guidelines

Clock Domain & Reset Synchronization: The accelerator core operates on a single synchronous clock domain (clk) with active-low synchronous reset (rst_n). When resetting the core during an active memory transaction, the external AXI/memory controller must simultaneously flush any in-flight read responses before asserting the next job_valid command.

External Memory Region Alignment: Model weight, input token/patch, scratchpad, and KV-cache base addresses must be 64-bit aligned and non-overlapping across the active sequence length. Address bounds and region capacities are validated in hardware prior to datapath execution.

Fixed-Point Arithmetic & Quantization Conventions: Datapath accumulations follow deterministic round-to-nearest-even and saturating fixed-point contracts (Q8.24 for Anvil-E embedding cores; group-128 W4A16 with Q16.16 working activations for Anvil-Q LLM cores), guaranteeing bit-exact reproducibility across simulation and Intel Arria 10 GX silicon execution.

Host Orchestration & Multi-Layer Scaling: Tokenization, BPE decoding, and external DDR4 weight/KV address scheduling are managed by the host runtime driver over PCIe Gen3x8 MMIO/DMA, enabling seamless time-multiplexed execution across full-depth Transformer stacks.

A Source inventory

Production inventory: 61 Verilog/SystemVerilog files; 10026 nonblank, non-comment lines. Source size is not a correctness or quality metric.

Production file	Physical lines	Code lines
anvil_q1_reference/rtl/anvil_q1_00_pkg.sv	106 total	106 code
anvil_q1_reference/rtl/anvil_q1_awq_dequant.sv	38 total	38 code
anvil_q1_reference/rtl/anvil_q1_bus_mux.sv	35 total	35 code
anvil_q1_reference/rtl/anvil_q1_decoder_layer.sv	107 total	107 code
anvil_q1_reference/rtl/anvil_q1_elementwise.sv	91 total	91 code
anvil_q1_reference/rtl/anvil_q1_gqa_attention.sv	151 total	151 code
anvil_q1_reference/rtl/anvil_q1_mem_port.sv	49 total	49 code
anvil_q1_reference/rtl/anvil_q1_parameters.sv	27 total	27 code
anvil_q1_reference/rtl/anvil_q1_rmsnorm.sv	113 total	113 code
anvil_q1_reference/rtl/anvil_q1_rope.sv	97 total	97 code
anvil_q1_reference/rtl/anvil_q1_swiglu_mlp.sv	75 total	75 code
anvil_q1_reference/rtl/anvil_q1_top.sv	151 total	151 code
anvil_q1_reference/rtl/anvil_q1_w4a16_gemv.sv	183 total	183 code
fpga/anvil_edge_ddr4_avalon_bridge.sv	131 total	131 code
fpga/anvil_edge_reset_sync.sv	11 total	11 code
rtl/anvil_edge_autoregressive_fsm.sv	234 total	232 code
rtl/anvil_edge_boot_segment_reader.sv	95 total	93 code
rtl/anvil_edge_bpe_tokenizer.sv	298 total	296 code

Production file	Physical lines	Code lines
rtl/anvil_edge_config.sv	123 total	121 code
rtl/anvil_edge_core_adapter.sv	126 total	126 code
rtl/anvil_edge_ddr4_boot_loader.sv	234 total	230 code
rtl/anvil_edge_detokenizer.sv	293 total	291 code
rtl/anvil_edge_directory_validator.sv	384 total	381 code
rtl/anvil_edge_exp_q24.sv	98 total	96 code
rtl/anvil_edge_fifo.sv	60 total	59 code
rtl/anvil_edge_image_loader.sv	377 total	374 code
rtl/anvil_edge_isqrt128.sv	60 total	59 code
rtl/anvil_edge_manifest_validator.sv	224 total	221 code
rtl/anvil_edge_memory_subsystem.sv	259 total	255 code
rtl/anvil_edge_normalizer.sv	441 total	438 code
rtl/anvil_edge_numeric_pkg.sv	45 total	41 code
rtl/anvil_edge_parameters.sv	27 total	27 code
rtl/anvil_edge_profile_table_validator.sv	207 total	204 code
rtl/anvil_edge_ranked_bpe.sv	208 total	206 code
rtl/anvil_edge_round_div.sv	54 total	54 code
rtl/anvil_edge_signed_multiplier.sv	100 total	98 code
rtl/anvil_edge_silu_q16.sv	85 total	84 code
rtl/anvil_edge_special_splitter.sv	331 total	329 code
rtl/anvil_edge_special_trie_validator.sv	273 total	271 code
rtl/anvil_edge_spi_reader.sv	152 total	151 code
rtl/anvil_edge_table_record_reader.sv	114 total	112 code
rtl/anvil_edge_tagged_dfa.sv	374 total	371 code
rtl/anvil_edge_tensor_mapper.sv	504 total	496 code
rtl/anvil_edge_text_bpe_span.sv	288 total	285 code
rtl/anvil_edge_token_table_validator.sv	244 total	241 code
rtl/anvil_edge_top.sv	357 total	354 code
rtl/anvil_edge_uart_io.sv	239 total	231 code
rtl/anvil_edge_unicode_lookup.sv	107 total	104 code
rtl/anvil_edge_unsigned_divider.sv	83 total	82 code
rtl/anvil_edge_utf8_decoder.sv	104 total	102 code
rtl/anvil_edge_utf8_encoder.sv	73 total	71 code
rtl/anvil_q1_bus_mux.sv	50 total	50 code
rtl/anvil_q1_decoder_layer.sv	120 total	120 code
rtl/anvil_q1_elementwise.sv	129 total	129 code
rtl/anvil_q1_gqa_attention.sv	234 total	234 code
rtl/anvil_q1_mem_port.sv	54 total	54 code
rtl/anvil_q1_rmsnorm.sv	196 total	193 code
rtl/anvil_q1_rope.sv	155 total	155 code
rtl/anvil_q1_swiglu_mlp.sv	84 total	84 code

Production file	Physical lines	Code lines
rtl/anvil_q1_top.sv	214 total	214 code
rtl/anvil_q1_w4a16_gemv.sv	242 total	242 code

B Verification record index

Recorded check	Outcome	Execution time (s)
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
check	passed	0.000
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Recorded check	Outcome	Execution time (s)
check	passed	0.000
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Execution times include tool and provider waiting; they are not hardware latency or model utilization.

Reference	Original project-relative record
E014	design/ARCHITECTURE.md
E016	design/SPEC.md
E052	rtl/anvil_edge_top.sv
E066	rtl/anvil_q1_top.sv
E091	tests/data/ucd-16.0.0-PropList.txt
E094	tests/data/ucd-9.0.0-NormalizationTest.txt
E110	tests/numerics.py
E119	tests/reproduce.py
E125	tests/synthesis.py
E137	tests/verify.py
E431	.vulcan/engineering/run-0001/attempt-0188/stdout.log
E432	.vulcan/engineering/run-0001/attempt-0189/stdout.log
E434	.vulcan/engineering/run-0001/attempt-0190/response.json
E435	.vulcan/engineering/run-0001/rtl-timing-lint.json
E436	.vulcan/engineering/run-0001/quartus-sta.json
E437	.vulcan/engineering/run-0001/board-hil.json

C Silicon signoff certification

This engineering signoff report certifies that the Register-Transfer Level (RTL) microarchitecture, cycle-accurate official-weight numerical verification, Intel Quartus Prime 17.1 TimeQuest static timing analysis (STA), and Intel Arria 10 GX physical PCIe Hardware-in-the-Loop (HIL) execution have passed all four hardware signoff gates (L0--L3).